

MODULE - 3 Beta & Gamma Function

Gamma Function ($\Gamma(n)$)

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$$

special function
 $e^{\infty} = \infty$
 $e^{-\infty} = 0$

FORMULAE:

$$\Gamma(n) = \begin{cases} (n-1)\Gamma(n-1) & \text{(for fractions)} \\ (n-1)! & \text{if } n \in \text{positive whole numbers} \end{cases}$$

Ex ① $\Gamma(10) = 9!$

$$\textcircled{1} \Gamma\left(\frac{3}{2}\right) = \frac{3}{2} \Gamma\left(\frac{3}{2}\right) \Rightarrow \frac{3}{2} \cdot \frac{1}{2} \Gamma\left(\frac{1}{2}\right) = \frac{3}{4} \sqrt{\pi}$$

$$\textcircled{3} \Gamma\left(\frac{10}{3}\right) = \frac{7}{3} \cdot \frac{4}{3} \cdot \frac{1}{3} \Gamma\left(\frac{1}{3}\right)$$

$$\textcircled{2} \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi} \quad \Gamma(1) = 1$$

$$\textcircled{3} \Gamma(0) = \Gamma(\text{negative integer}) = \infty \quad \text{Not defined.}$$

$$\textcircled{4} \Gamma(p) \Gamma(1-p) = \frac{\pi}{\sin p\pi}, \quad 0 < p < 1$$

Gamma functions are not defined for negative integers. But it is defined for negative fractions.

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$$\sqrt{\frac{1}{4}} \sqrt{\frac{3}{4}} = \frac{\pi}{\sin(\frac{\pi}{4})} \rightarrow \underline{\underline{\sqrt{2}\pi}}$$

$$\sqrt{\frac{1}{6}} \sqrt{\frac{5}{6}} = \frac{\pi}{\sin(\frac{\pi}{6})} = \underline{\underline{2\pi}}$$

$$\sqrt{\frac{1}{3}} \sqrt{\frac{2}{3}} = \frac{\pi}{\sin(\frac{\pi}{3})} = \underline{\underline{\frac{2\pi}{\sqrt{3}}}}$$

Evaluate:

$$1) \int_0^{\infty} \sqrt[4]{x} e^{-\sqrt{x}} dx \quad I = \int_0^{\infty} x^{1/4} e^{-(\sqrt{x})} dx$$

$$I_n = \int_0^{\infty} e^{-x^2} x^n dx$$

$$\text{put } \sqrt{x} = t \rightarrow x = t^2 \\ dx = 2t dt$$

$$I = \int_{t=0}^{\infty} (t^2)^{1/4} e^{-t} (2t) dt$$

$$2 \int_0^{\infty} e^{-t} t^{(3/2+1)-1} dt \rightarrow 2 \int_0^{\infty} e^{-t} t^{5/2-1} dt$$

$$2 \sqrt{\frac{3}{2}} \quad I_n = (n-1) I_{n-1}$$

$$2 \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}} \rightarrow \underline{\underline{\frac{3}{2} \sqrt{\pi}}}$$

if you see:

$$t^a e^{-t}$$

$$\int_0^{\infty} t^a e^{-t} dt$$

$$= \Gamma(a+1)$$

$$\therefore a = n-1$$

$$\underline{\underline{2}} \quad \int_0^{\infty} (x^2 + 4) e^{-x^2} dx$$

$$\text{let } I = \int_0^{\infty} (x^2 + 4) e^{-x^2} dx$$

$$2x^2 = t \rightarrow x = \frac{\sqrt{t}}{\sqrt{2}} \rightarrow dx = \frac{1}{2\sqrt{2}} \cdot \frac{1}{\sqrt{t}} dt$$

$$\begin{array}{c|c|c} x & 0 & \infty \\ \hline t & 0 & \infty \end{array}$$

$$I = \int_0^{\infty} \left(\frac{t}{2} + 4 \right) e^{-t} \cdot \frac{1}{2\sqrt{2}} t^{-1/2} dt$$

$$\frac{1}{2\sqrt{2}} \int_0^{\infty} \left\{ e^{-t} \frac{t^{1/2}}{2} + 4 e^{-t} t^{-1/2} \right\} dt$$

$$\frac{1}{2\sqrt{2}} \left\{ \frac{1}{2} \int_0^{\infty} e^{-t} t^{3/2-1} dt + 4 \int_0^{\infty} e^{-t} t^{1/2-1} dt \right\}$$

$$\frac{1}{2\sqrt{2}} \left\{ \frac{1}{2} \sqrt{\frac{3}{2}} + 4 \sqrt{\frac{1}{2}} \right\}$$

$$\frac{1}{2\sqrt{2}} \left\{ \frac{1}{2} \cdot \frac{1}{2} \sqrt{\frac{3}{2}} + 4 \sqrt{\frac{1}{2}} \right\}$$

$$\frac{1}{2\sqrt{2}} \left\{ \frac{\sqrt{\pi}}{4} + 4\sqrt{\pi} \right\} = \frac{17\sqrt{\pi}}{8\sqrt{2}}$$

Show That

$$\textcircled{1} \int_0^{\infty} x e^{-x^8} dx \times \int_0^{\infty} x^2 e^{-x^4} dx = \frac{\pi}{16\sqrt{2}}$$

$$\textcircled{2} \int_0^{\infty} \sqrt{y} e^{-y^2} dy \times \int_0^{\infty} \frac{e^{-y^2}}{\sqrt{y}} dy = \frac{\pi}{2\sqrt{2}}$$

$$\textcircled{1} I = I_1 \times I_2$$

$$I_1 = \int_0^{\infty} x e^{-x^8} dx$$

$$x^8 = t \rightarrow x = t^{1/8}$$

$$dx = \frac{1}{8} t^{-7/8} dt$$

$$I_2 = \int_0^{\infty} x^2 e^{-x^4} dx$$

$$x^4 = u \rightarrow x = u^{1/4}$$

$$dx = \frac{1}{4} u^{-3/4} du$$

x	0	∞
t	0	∞

x	0	∞
u	0	∞

$$I_1 = \int_0^{\infty} t^{1/8} e^{-t} \cdot \frac{1}{8} t^{-7/8} dt$$

$$\int_0^{\infty} x^m e^{-x^n} dx$$

↓

$$\frac{1}{8} \int_0^{\infty} e^{-t} t^{-3/4} dt = \frac{1}{8} \int_0^{\infty} e^{-t} t^{1/4-1} dt$$

$$\frac{1}{n} \Gamma\left(\frac{m+1}{n}\right)$$

$$I_1 = \frac{1}{8} \Gamma\left(\frac{1}{4}\right) \textcircled{1}$$

$$I_1 = \frac{1}{8} \sqrt{\frac{1}{4}} \quad \text{--- ①}$$

$$I_2 = \int_0^{\infty} (u^{1/4})^2 e^{-u} \frac{1}{4} u^{-3/4} du$$

$$= \frac{1}{4} \int_0^{\infty} e^{-u} u^{(-1/4+1)} du = \frac{1}{4} \int_0^{\infty} e^{-u} u^{3/4} du$$

$$I_2 = \frac{1}{4} \sqrt{\frac{3}{4}} \quad \text{--- ②}$$

$$\frac{ST}{\Gamma} \int_0^{\infty} x e^{-x^2} dx \int_0^{\infty} x^2 e^{-x^4} dx = \frac{\pi}{16\sqrt{2}} = 8k$$

$$I = \left(\frac{1}{8} \sqrt{\frac{1}{4}} \right) \left(\frac{1}{4} \sqrt{\frac{3}{4}} \right)$$

$$\frac{1}{32} \left(\sqrt{\frac{1}{4}} \sqrt{\frac{3}{4}} \right)$$

$$|p| |1-p| = \frac{\pi}{\sin \pi} \quad (0 < p < 1)$$

$$I = \frac{1}{32} \frac{\pi}{\left(\sin \frac{\pi}{4} \right)} = \frac{\sqrt{2} \pi}{32} \Rightarrow \frac{\pi}{16\sqrt{2}}$$

Type-2

$$\left. \begin{aligned} \log x = -t &\rightarrow x = e^{-t} \\ \log \frac{1}{x} = t &\rightarrow \frac{1}{x} = e^t \rightarrow x = e^{-t} \end{aligned} \right\}$$

$$\begin{aligned} \log 1 &= 0 \rightarrow e^0 = 1 \\ \log \infty &= \infty \rightarrow e^\infty = \infty \\ \log 0 &= -\infty \rightarrow e^{-\infty} = 0 \end{aligned}$$

$$\text{S.T. } \int_0^1 (x \log x)^3 dx = \frac{-3}{128}$$

$$\text{let } I = \int_0^1 x^3 (\log x)^3 dx$$

$$\text{put } \log x = -t \quad \boxed{\log x = -t}$$

$$Q. \text{ Show that } \int_0^1 (x \log x)^3 dx = \frac{-3}{128}$$

$$\text{let } I = \int_0^1 x^3 (\log x)^3 dx$$

$$\text{put } \log x = -t$$

$$x = e^{-t} \rightarrow dx = -e^{-t} dt$$

$$x=0 \rightarrow \log 0 = -t \quad \left| \quad x=1 \rightarrow \log 1 = -t \right.$$

$$-\infty = -t$$

$$0 = -t$$

$$t = \infty$$

$$t = 0$$

$$x=1 \rightarrow \log 1 = -t$$

TRICK FORMULA: $\int x^m \ln x$

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$$\frac{x}{t} \bigg|_{0/\infty}$$

$$I = \int_0^\infty (e^{-t})^3 (-t)^3 (-e^{-t} dt)$$

$$\rightarrow (-t)^3 (-1) \int_0^\infty e^{-4t} \cdot t^3 dt$$

$$\rightarrow t \int_0^\infty e^{-4t} \cdot t^3 dt$$

$$\text{put } 4t = u \rightarrow t = \frac{u}{4}$$

$$\frac{t}{u} \bigg|_{\infty/0}$$

$$dt = \frac{du}{4}$$

$$I = \int_0^\infty (e^{-t})^3 (-t)^3 (-e^{-t} dt)$$

$$(-1)^3 (-1) \int_0^\infty e^{-4t} \cdot t^3 dt$$

$$\text{put } 4t = u \rightarrow t = \frac{u}{4}$$

$$\frac{t}{u} \bigg|_{\infty/0}$$

$$I = \int_0^\infty e^{-4} \left(\frac{u}{4}\right)^3 \frac{du}{4}$$

$$= \frac{1}{(4)^4} \int_0^\infty e^{-4} u^{4-1} du$$

$$= \frac{1}{(4)^4} \left[\frac{e^{-4}}{4} \right]_0^\infty = (-1) \frac{1 \times 2 \times 3}{4^4} = \frac{-6}{4^4}$$

$$\int_0^1 x^m (\log x)^n dx = (-1)^n \frac{n!}{(m+1)^{n+1}}$$

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$$\frac{-63}{24 \times 4 \times 16}$$

$$\frac{-3}{128}$$

Formula used

$$\int_0^1 x^m (\log x)^n dx = (-1)^n \frac{n!}{(m+1)^{n+1}}$$

$$(x \log x)^3 dx$$

$$\int_0^1 (x^3) (\log x)^3 dx \rightarrow n=3, m=3.$$

$$\frac{(-1)^3 \cdot 3!}{(4)^4} \rightarrow \frac{-1 \cdot 3 \times 2}{24 \times 4 \times 16} \rightarrow \frac{-1 \cdot 3}{8 \times 16} = \frac{-3}{128}$$

Show that

$$1) \int_0^1 x^m (\log x)^n dx = \frac{(-1)^n n!}{(m+1)^{n+1}}$$

$$2) \int_0^1 x^m \left(\log \frac{1}{x}\right)^n dx = \frac{n!}{(m+1)^{n+1}}$$

$$3) \int_0^1 \frac{dx}{\sqrt{x \log x}} = \sqrt{2\pi}$$

② let $I = \int_0^1 x^m \left(\log \frac{1}{x}\right)^n dx$

put $\log \frac{1}{x} = t \Rightarrow x = e^{-t}$
 $dx = -e^{-t} dt$

t	0	∞
x	1	0

$x=0 \rightarrow \log \infty = t$
 $x=1 \rightarrow \log 1 = t$
 $t=0$

$$I = \int_0^{\infty} e^{-mt} t^n (-e^{-t} dt)$$

$$I = \int_0^{\infty} e^{-u} \left(\frac{u}{m+1} \right)^n \frac{du}{m+1}$$

$$= - \int_0^{\infty} e^{-(m+1)t} t^n dt$$

Show that

$$1) \int_0^{\infty} \frac{x^a}{a^x} dx = \frac{\Gamma(a+1)}{(\log a)^{a+1}} \quad (a > 1)$$

Type-3

$$2) \int_0^{\infty} 7^{-4x^2} dx = \frac{\sqrt{\pi}}{4\sqrt{\log 7}}$$

Type-4

Q let $I = \int_0^{\infty} a^{-x} x^a dx$

we want $\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$

put $a^{-x} = e^{-x}$

$-x \log a = -t$

$$\boxed{x = \frac{t}{\log a}}$$

$dx = \frac{dt}{\log a}$

x	0	∞
t	0	∞

$$I = \int_0^{\infty} e^{-t} \left(\frac{t}{\log a} \right)^a \frac{dt}{\log a}$$

$$\frac{1}{(\log a)^{a+1}} \int_0^{\infty} e^{-t} t^{(a+1)-1} dt$$

instead of t^a we add
 & subtracted 1 to bring
 it in gamma function

$$\frac{1}{(\log a)^{a+1}} \int_0^{\infty} e^{-t} t^a dt$$

$$2) \int_0^{\infty} 7^{-4x^2} dx = \frac{\sqrt{\pi}}{4\sqrt{\log 7}}$$

put $7^{-4x^2} = e^{-t}$

take log on both sides.

$$7 \cdot 4x^2 \log 7 = -t$$

$$x^2 = \frac{-t}{4 \log 7}$$

$$dx = \frac{1}{2\sqrt{\log 7}} \cdot \frac{1}{2\sqrt{-t}} dt$$

$$x = \frac{\sqrt{-t}}{2\sqrt{\log 7}}$$

x	0	∞
t	0	∞

$$\rightarrow I = \int_0^{\infty} e^{-t} \frac{1}{2\sqrt{\log 7}} \frac{1}{2\sqrt{-t}} dt$$

$$I = \int_0^{\infty} e^{-t} \frac{1}{4\sqrt{\log 7}} t^{-1/2} dt$$

$$I = \frac{1}{4\sqrt{\log 7}} \int_0^{\infty} e^{-t} t^{-1/2} dt$$

$$= \frac{1}{4\sqrt{\log 7}} \Gamma\left(\frac{1}{2}\right)$$

BETA FUNCTION $B(m, n)$

$$1) B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

definition.

Properties & Formulae:

$$2) B(m, n) = B(n, m)$$

$$3) B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$4) B(m, n) = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin^{2m-1} \theta \cos^{2n-1} \theta d\theta$$

put $p = 2m-1, q = 2n-1$

$$m = \frac{p+1}{2}, n = \frac{q+1}{2}$$

$$5) B\left(\frac{p+1}{2}, \frac{q+1}{2}\right) = \frac{\pi}{2} \int_0^{\frac{\pi}{2}} \sin^p \theta \cos^q \theta d\theta$$

Ex $\int_0^{\frac{\pi}{2}} \sin^{\infty} \theta \cos^{\infty} \theta d\theta = \frac{1}{2} B\left(\frac{101}{2}, \frac{101}{2}\right)$

$$⑥ \quad B(m, n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

Show that $\int_0^{2a} x^2 \sqrt{2ax - x^2} dx = \frac{8\pi a^4}{3}$

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$I = \int_0^{2a} x^2 \sqrt{x(2a-x)} dx$$

$$= \int_0^{2a} x^{3/2} (2a-x)^{1/2} dx$$

put $x = 2at \rightarrow dx = (2a) dt$

x	0	$2a$
t	0	1

$$I = \int_0^1 (2at)^{3/2} (2a-2at)^{1/2} (2a) dt$$

$$= (2a)^4 \int_0^1 t^{3/2} (1-t)^{1/2} dt$$

$$(2a)^4 \int_0^1 t^{7/2-1} (1-t)^{3/2-1} dt$$

$$(2a)^4 B\left(\frac{7}{2}, \frac{3}{2}\right)$$

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$= \frac{(2a)^4}{15} \sqrt{\frac{7}{2}} \sqrt{\frac{3}{2}}$$

$$= \frac{8\pi a^4}{3}$$

$$\frac{(2a)^4}{4!} \cdot \frac{3}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$$

$$\frac{\pi a^4}{8}$$

Show that

$$\textcircled{1} \int_0^a x^4 (a^2 - x^2)^{1/2} dx = \frac{\pi a^6}{32}$$

$$\textcircled{2} \int_0^1 \frac{x^7}{\sqrt{1-x^2}} dx = \frac{16}{35}$$

$$\textcircled{3} \int_0^1 \frac{x^2(4-x^4)}{\sqrt{1-x^2}} dx = \frac{27\pi}{32}$$

$$\textcircled{4} \int_0^2 x \sqrt{8-x^3} dx = \frac{16\pi}{9\sqrt{3}}$$

Ans $\textcircled{1} \int_0^a x^4 (a^2 - x^2)^{1/2} dx = \frac{\pi a^6}{32}$

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Put $x^2 = at$ $\rightarrow x = a\sqrt{t}$ $dx = \frac{a}{2\sqrt{t}} dt$

$$x=0 \rightarrow t=0$$

$$x=a \rightarrow t=1$$

$$I = \int_0^1 (a^2 t)^2 (a^2 - a^2 t)^{1/2} \cdot \frac{a}{2\sqrt{x}} dt$$

$$= \frac{a^4 \cdot a \cdot a}{2} \cdot \frac{a^6}{2} \int_0^1 t^{3/2} (1-t)^{1/2} dt$$

$$= \frac{a^6}{2} \int_0^1 t^{3/2-1} (1-t)^{1/2-1} dt$$

$$\frac{a^6}{2} B\left(\frac{5}{2}, \frac{3}{2}\right)$$

$$B(m, n) = \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)}$$

$$\frac{a^6}{2} \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{3}{2}\right)}{\Gamma(4)}$$

$$\frac{a^6}{2 \times 3!} \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}} \quad \left| \Gamma(n) = (n-1) \Gamma(n-1) \right.$$

$$\frac{a^6}{2(3!)} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\frac{1}{2}} \cdot \frac{1}{2} \sqrt{\frac{1}{2}} \rightarrow \frac{a^6 \pi}{32 \sqrt{2}}$$

$$(4) \int_0^2 x \sqrt[3]{8-x^3} dx \rightarrow \int_0^2 x (8-x^3)^{1/3} dx$$

$$B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

$$\text{put } x^3 = 8t \rightarrow x = 2t^{1/3} \rightarrow dx = \frac{2}{3} t^{-2/3} dt$$

$$x=0 \rightarrow t=0 \rightarrow I = \int_0^1 2t^{1/3} (8-8t)^{1/3} \cdot \frac{2}{3} t^{-2/3} dt$$

$$x=2 \rightarrow t=1$$

$$\rightarrow \frac{8}{3} \int_0^1 t^{-1/3} (1-t)^{1/3} dt$$

$$\frac{8}{3} \int_0^1 t^{2/3-1} (1-t)^{4/3-1} dt$$

$$\frac{8}{3} B\left(\frac{2}{3}, \frac{4}{3}\right)$$

$$\frac{8}{3} \frac{\sqrt{\frac{2}{3}} \sqrt{\frac{4}{3}}}{\sqrt{\frac{1}{3}}} \rightarrow \frac{8}{3} \frac{\sqrt{\frac{2}{3}} \sqrt{\frac{4}{3}} \sqrt{\frac{12}{3}}}{\sqrt{\frac{1}{3}}}$$

$$\frac{8}{3(1!)} \sqrt{\frac{2}{3}} \frac{1}{3} \sqrt{\frac{1}{3}} \quad \left| \quad \Gamma(p) \Gamma(1-p) = \frac{\pi}{\sin p\pi} \right.$$

$$\frac{8}{9} \sqrt{\frac{10}{3}} \sqrt{\frac{2}{3}}$$

$$\frac{8}{9} \frac{\Gamma \frac{2}{3}}{\sin \frac{\pi}{3}} \rightarrow \frac{8\pi \frac{2}{3}}{9 \frac{\sqrt{3}}{2}} \rightarrow \frac{16\pi}{9\sqrt{3}}$$

Show that.

$$\textcircled{1} \int_0^1 \sqrt{1-x} dx \quad \int_0^{1/2} \sqrt{2y+y^2} dy = \frac{\pi}{30}$$

$$\textcircled{2} \int_0^1 \frac{x^3}{\sqrt{1-x^4}} dx \quad \int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\pi}{4}$$

$$\textcircled{3} \int_0^1 \frac{dx}{\sqrt{1-x^{2n}}} = \sqrt{1+\frac{1}{n}} \sqrt{1-\frac{1}{n}}$$

$$\underline{\underline{2}} \int_0^1 \frac{x^2}{\sqrt{1-x^4}} dx = \int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\pi}{4}$$

$$I = \int_0^1 x^2 (1-x^4)^{-1/2} dx = \int_0^1 (1-x^4)^{-1/2} dx.$$

$$\text{Put } x^4 = t \rightarrow x = t^{1/4} \quad dx = \frac{1}{4} t^{-3/4} dt.$$

$$x=0 \rightarrow t=0 \quad x=1 \rightarrow t=1$$

$$I = \int_0^1 t^{1/2} (1-t)^{-1/2} \cdot \frac{1}{4} t^{-3/4} dt$$

$$I = \int_0^1 t^{1/2} (1-t)^{-1/2} \cdot \frac{1}{4} t^{-3/4} dt = \frac{1}{4} \int_0^1 t^{-1/4} (1-t)^{-1/2} dt$$

$$= \frac{1}{16} \int_0^1 t^{-1/4} (1-t)^{-1/2} dt = \frac{1}{16} \int_0^1 t^{-3/4} (1-t)^{-1/2} dt$$

$$= \frac{1}{16} \int_0^1 t^{3/4-1} (1-t)^{1/2-1} dt = \frac{1}{16} \int_0^1 t^{1/4-1} (1-t)^{1/2-1} dt$$

$$= \frac{1}{16} B\left(\frac{3}{4}, \frac{1}{2}\right) \cdot B\left(\frac{1}{4}, \frac{1}{2}\right)$$

$$= \frac{1}{16} \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{5}{4}\right)} \cdot \frac{\Gamma\left(\frac{1}{4}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{3}{4}\right)} = \frac{\pi}{16} \frac{\Gamma\left(\frac{3}{4}\right) \Gamma\left(\frac{1}{4}\right)}{\Gamma\left(\frac{5}{4}\right) \Gamma\left(\frac{3}{4}\right)}$$

$$\underline{\underline{\frac{\pi}{4}}}$$

②

$$\int \frac{dx}{x^{1-n}} = \left(1+\frac{1}{n}\right) \left(1-\frac{1}{n}\right)$$

$$\int (1-x^n)^k dx \rightarrow x^n = t$$

$$x = (t)^{1/n}$$

$$dx = \frac{1}{n} (t)^{\frac{1}{n}-1} dt$$

$$\frac{1}{n} (t)^{\frac{1}{n}-1} \left(1-t\right)^k \cdot \frac{1}{n} (t)^{\frac{1}{n}-1} dt$$

$$2 = \int_0^1 (1-t)^{1/n} \cdot \frac{1}{n} (t)^{\frac{1}{n}-1} dt$$

$$\frac{1}{n} \int_0^1 t^{\left(\frac{1}{n}-1\right)} \cdot (1-t)^{\left(\frac{1}{n}+1\right)-1} dt$$

$$\frac{1}{n} B\left(\frac{1}{n}, 1-\frac{1}{n}\right)$$

$$\frac{1}{n} \frac{\Gamma\left(\frac{1}{n}\right) \Gamma\left(1-\frac{1}{n}\right)}{\Gamma\left(\frac{1}{n} + 1 - \frac{1}{n}\right)}$$

$$\frac{1}{n} \frac{\Gamma\left(\frac{1}{n}\right) \Gamma\left(1-\frac{1}{n}\right)}{\Gamma(1)}$$

$$\frac{1}{n} \frac{\Gamma\left(\frac{1}{n}\right) \Gamma\left(1-\frac{1}{n}\right)}{\Gamma(1)}$$

HW

①

$$\int_0^1 \frac{x^7}{\sqrt{1-x^2}} dx$$

$$\int_0^1 x^7 (1-x^2)^{-1/2} dx$$

$$\int_0^1 x^{m-1} (1-x)^{n-1} dx$$

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$$x^2 = t$$

$$dt = 2x dx$$

$$dx = \frac{dt}{2x}$$

$$x^7 = (x^2)^3 \cdot x$$

$$x^7 = t^3 x$$

$$x^7 dx = \frac{t^3 dt}{2}$$

$$(1-x^2)^{-1/2} \rightarrow (1-t)^{-1/2}$$

$$\frac{1}{2} \int_0^1 t^3 (1-t)^{-1/2} dt \rightarrow \frac{1}{2} \int_0^1 t^{m-1} (1-t)^{n-1} dt$$

$$m=4, n=1/2. B(4, 1/2) \rightarrow \frac{4! \sqrt{\frac{1}{2}}}{3! \sqrt{\pi}}$$

$$\frac{7}{2} \times \frac{5}{2} \times \frac{3}{2} \times \frac{1}{2} \sqrt{\pi} \rightarrow \frac{1 \cdot 3 \times 2 \times 2 \times 2 \times 2}{7 \times 5 \times 3 \times 1}$$

$$\frac{16}{35} \quad B(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx$$

Evaluate $\int_0^{\pi/2} x^5 \sin^7 x dx$

$$1) \int_0^{\pi/2} x^4 \cos^6 x dx \quad I = \int_0^{\pi/2} x^4 \cos^6 x dx$$

$$\int u v dx = u \int v dx - \int \frac{du}{dx} \int v dx dx$$

$$I = \int_0^{\pi/2} (\cos^6 x) x^4 dx$$

$$\left[\frac{\cos^6 x x^5}{5} \right]_0^{\pi/2} - \int_0^{\pi/2} \frac{-1}{\sqrt{1-x^2}} \cdot \frac{x^5}{5} dx$$

$$= (0-0) + \frac{1}{5} \int_0^{\pi/2} x^5 (1-x^2)^{-1/2} dx$$

$$B(3, 1/2)$$

$$\frac{3!}{3!} \frac{1/2!}{1/2!}$$

$$\rightarrow \frac{2!}{2!} \frac{1!}{1!}$$

$$\frac{7!}{2!}$$

$$\frac{5!}{2!} \frac{3!}{2!} \frac{1!}{2!}$$

$$10 \rightarrow \frac{1}{5} \int_0^1 x^3 (1-x^2)^{-1/2} dx$$

$$x = \sqrt{t} \rightarrow dx = \frac{1}{2\sqrt{t}} dt$$

$$\frac{x}{t} \left| \frac{0}{0} \right| \frac{1}{1}$$

$$I = \frac{1}{5} \int_0^1 t^{3/2} (1-t)^{-1/2} \frac{1}{2} t^{-1/2} dt$$

$$\frac{1}{10} \int_0^1 t^2 (1-t)^{-1/2} dt$$

$$I = \frac{1}{10} \int_0^1 t^{3-1} (1-t)^{1/2-1} dt \rightarrow \frac{1}{10} B(3, \frac{1}{2})$$

$$\frac{1}{10} \frac{3!}{3!} \frac{1/2!}{1/2!}$$

$$\rightarrow \frac{2!}{2!} \frac{1!}{1!}$$

$$\frac{5!}{2!} \frac{3!}{2!} \frac{1!}{2!}$$

$$\times \frac{1}{10}$$

$$\frac{2 \times 2 \times 2 \times 2}{15}$$

$$\times \frac{1}{10}$$

$$\frac{8}{75}$$

Show that

$$\textcircled{1} \int_0^\pi \frac{\sin^4 \theta}{(1+\cos \theta)^2} d\theta = \frac{3\pi}{2}$$

$$\textcircled{2} \int_0^\pi \tan \theta d\theta \times \int_0^\pi \sqrt{\cot \theta} d\theta = \frac{\pi^2}{2}$$

$$(3) \int_0^{\pi} \frac{\sin^4 \theta}{\sqrt{\cos \theta}} d\theta = \frac{1}{7} \sqrt{\frac{2}{\pi}} \left(\sqrt{\frac{1}{4}} \right)^2$$

$$(4) \int_0^{\pi} \sin^2 6\theta \cos^6 3\theta d\theta = \frac{7\pi}{384}$$

Ans 1) $\int_0^{\pi} \frac{\sin^4 \theta}{(1+\cos \theta)^2} d\theta = \frac{3\pi}{2}$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$1 + \cos 2\theta = 2 \cos^2 \theta$$

$$\int_0^{\pi} \frac{(2 \sin \theta / 2 \cos \theta / 2)^4}{(2 \cos^2 \theta / 2)^2} d\theta$$

$$\theta = 2t \rightarrow d\theta = 2dt$$

$$= 2^2 \int_0^{\pi} \sin^4 (\theta/2) d\theta$$

$$\begin{array}{c|c|c} \theta & 0 & \pi \\ \hline t & 0 & \pi/2 \end{array}$$

$$= 4 \int_0^{\pi/2} \sin^4 t \cos^0 t (2dt)$$

$$8 \int_0^{\pi/2} \sin^4 t \cos^0 t dt \rightarrow$$

$$\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta = \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$= 8 \cdot \frac{1}{2} B\left(\frac{4+1}{2}, \frac{0+1}{2}\right) \rightarrow 4 B\left(\frac{5}{2}, \frac{1}{2}\right)$$

$$B(m, n) \rightarrow \frac{\Gamma(m) \Gamma(n)}{\Gamma(m+n)} \rightarrow \frac{\Gamma\left(\frac{5}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma(3)} \rightarrow \frac{\frac{3}{2} \cdot \frac{1}{2} \pi \cdot \pi}{2}$$

$$\frac{\frac{3}{4} \pi}{2} \rightarrow \frac{3 \times 84 \pi}{2} \rightarrow \frac{3\pi}{2}$$

3. $I = \int_0^{\pi} \sin^4 \theta \cos^{12} \theta d\theta$
 $= \frac{1}{2} B\left(\frac{4+1}{2}, \frac{12+1}{2}\right)$

$\frac{1}{2} B\left(\frac{5}{2}, \frac{13}{2}\right)$

$\frac{5+1}{2} \rightarrow \frac{10+1}{2}$

$\frac{13}{2}$

$\frac{\sqrt{\frac{5}{2}} \sqrt{\frac{13}{2}}}{\sqrt{\frac{11}{2}}} \rightarrow \frac{\cancel{\frac{3}{2}} \cdot \cancel{\frac{1}{2}} \cdot \sqrt{\frac{1}{2}} \sqrt{\frac{1}{2}}}{\frac{7}{2} \cdot \frac{3}{2} \sqrt{\frac{3}{2}}}$

~~$\frac{1}{2} \frac{\sqrt{\frac{5}{2}} \sqrt{\frac{13}{2}}}{\sqrt{\frac{11}{2}}}$~~

$\frac{1}{2} \frac{\sqrt{\frac{5}{2}} \sqrt{\frac{13}{2}}}{\sqrt{\frac{11}{2}}} \rightarrow \frac{1}{2} \left(\frac{\cancel{\frac{3}{2}} \cdot \cancel{\frac{1}{2}} \cdot \sqrt{\frac{1}{2}}\right) \left(\sqrt{\frac{1}{2}}\right) \sqrt{\frac{1}{2}}}{\frac{7}{2} \cdot \frac{3}{2} \cdot \sqrt{\frac{3}{2}} \cdot \sqrt{\frac{3}{2}}}$

$\frac{1}{2} \times \frac{1 \cdot 2}{7} \sqrt{\frac{1}{2}} \rightarrow \frac{2}{7} \sqrt{\frac{1}{2}} \times \frac{1}{\sqrt{\frac{3}{2}}} \times \frac{1}{\sqrt{\frac{3}{2}}}$

where $\frac{\sqrt{\frac{1}{2}} \sqrt{\frac{3}{2}}}{\sin \frac{\pi}{4}} = \underline{\underline{\sqrt{2} \pi}}$

$$I = \frac{2\sqrt{\pi}}{7} \left(\sqrt{\frac{1}{4}} \right)^2 \Rightarrow \frac{1}{7} \sqrt{\frac{2}{\pi}} \left(\sqrt{\frac{1}{4}} \right)^2$$

Show that

$$\textcircled{1} \int_0^{\infty} \frac{x^2}{(1+x^6)^3} dx = \frac{\pi}{16} \quad \textcircled{2} \int_0^{\infty} \left(\frac{x}{1+x^2} \right)^6 dx = \frac{3\pi}{32}$$

$$\textcircled{3} \int_0^{\infty} \frac{dy}{1+y^4} = \frac{\pi}{2\sqrt{2}} \quad \textcircled{4} \int_0^{\infty} \frac{x^4(1-x^6)}{(1+x^2)^{24}} dx = 0$$

$$\textcircled{1} \text{ let } I = \int_0^{\infty} \frac{x^2}{(1+x^6)^3} dx \quad B(m,n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$x^6 = t \Rightarrow x = t^{1/6} \Rightarrow dx = \frac{1}{6} t^{-5/6} dt$$

$$\frac{x}{t} \mid \begin{matrix} 0 & \infty \\ t & 0 & \infty \end{matrix} \quad I = \int_0^{\infty} \frac{(t^{1/6})^2}{(1+t)^3} \cdot \frac{1}{6} t^{-5/6} dt$$

$$\frac{1}{6} \int_0^{\infty} \frac{t^{-1/2}}{(1+t)^3} dt \rightarrow \frac{1}{6} \int_0^{\infty} \frac{t^{1/2-1}}{(1+t)^{1/2+3/2}} dt \quad \begin{matrix} m = \frac{1}{2} \\ m+n = 3 \\ n = \frac{5}{2} \end{matrix}$$

$$\frac{1}{6} B\left(\frac{1}{2}, \frac{5}{2}\right) = \frac{1}{6} \frac{\Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{5}{2}\right)}{\Gamma(3)} \rightarrow \frac{1}{6(2!)} \sqrt{\pi} \cdot \frac{3}{2} \cdot \frac{1}{2} \sqrt{\pi}$$

$$(3) \int_0^{\infty} \frac{dy}{(1+y^2)^{3/2}} = \frac{\pi}{2\sqrt{2}}$$

$$y^4 = t \Rightarrow y = t^{1/4} \\ dy = \frac{1}{4} t^{-3/4} dt$$

$$\frac{y}{1+y^2} \Big|_0^{\infty}$$

$$I = \int_0^{\infty} \frac{1}{(1+t)^{3/2}} \cdot \frac{1}{4} t^{-3/4} dt$$

$$\frac{1}{4} \int_0^{\infty} \frac{t^{1/4-1}}{(1+t)^{3/2+1/4}} dt = \frac{1}{4} B\left(\frac{1}{4}, \frac{3}{4}\right)$$

$$m = \frac{1}{4} \quad m+n=1 \Rightarrow n = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\frac{1}{4} \sqrt{\frac{2}{\pi}} \sqrt{\frac{3}{\pi}} = \frac{1}{4} \frac{\pi}{\sqrt{2}}$$

$$\frac{1}{4} \frac{\pi}{\sqrt{2}} \Rightarrow \frac{\pi}{4\sqrt{2}}$$

$$\frac{1}{4} \frac{\pi}{\sqrt{2}}$$

$$\frac{1}{4} \frac{\pi}{\sqrt{2}}$$

$$\frac{1}{4} \frac{\pi}{\sqrt{2}} = \frac{\pi}{4\sqrt{2}}$$

$$\text{ST} \int_0^{\infty} \frac{x^{m-1}}{(a+bx)^{m+n}} dx = \frac{B(m,n)}{a^n b^m} \text{ \& hence evaluate}$$

$$\int_0^{\infty} \frac{\sqrt{x}}{(1+4x+x^2)} dx$$

$$B(m,n) = \int_0^{\infty} \frac{x^{m-1}}{(1+x)^{m+n}} dx$$

$$I = \int_0^{\infty} \frac{x^{m-1}}{(a+bx)^{m+n}} dx \rightarrow \text{put } bx=at \rightarrow \frac{x}{a+bx} \rightarrow \frac{x|_0^{\infty}}{a+bx|_0^{\infty}}$$

$$dx = \frac{a dt}{b}$$

$$\rightarrow I = \int_0^{\infty} \frac{\left(\frac{at}{b}\right)^{m-1}}{(a+at)^{m+n}} \frac{a dt}{b} = \frac{a^m}{b^m a^{m+n}} \int_0^{\infty} \frac{t^{m-1}}{(1+t)^{m+n}} dt$$

$$\frac{1}{b^m a^n} \int_0^{\infty} \frac{t^{m-1}}{(1+t)^{m+n}} dt \rightarrow \frac{1}{a^n b^m} B(m,n)$$

$$\text{let } I_1 = \int_0^{\infty} \frac{\sqrt{x}}{4+4x+x^2} dx \rightarrow \int_0^{\infty} \frac{x^{3/2-1}}{(2+x)^{3/2+1/2}} dx$$

$$m=3/2 \quad n=1/2 \quad m+n=2 \quad a=2 \quad b=1$$

$$\int_0^{\infty} \frac{x^{3/2-1}}{(2+x)^{3/2+1/2}} dx$$

$$\therefore \int_0^{\infty} \frac{x^{m-1}}{(a+bx)^{m+n}} dx$$

$$I_1 = \frac{B(m,n)}{a^n b^m} = \frac{B(3/2, 1/2)}{(2)^{3/2} (1)^{1/2}} \rightarrow \frac{1}{2\sqrt{2}} \frac{\Gamma(3/2) \Gamma(1/2)}{\Gamma(2)}$$

$$\rightarrow \frac{\pi}{2\sqrt{2}}$$

Duplication Formula

$$\Gamma(n) \Gamma\left(n + \frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2n-1}} \Gamma(2n)$$

Show that ① $\int_0^1 \frac{dx}{\sqrt{1-x^{2n}}} = \frac{2^{2n-1}}{n} \left(\frac{\sqrt{\pi}}{2}\right)^2$

② $\int_0^1 \frac{x^{2n}}{\sqrt{1-x^2}} dx = \frac{(2n)! \pi}{2^{2n+1} (n!)^2}$

③ $B(m, m) B\left(m + \frac{1}{2}, m + \frac{1}{2}\right) = \frac{\pi}{2} 2^{1-4m}$

Ans ① let $I = \int_0^1 \frac{dx}{\sqrt{1-x^{2n}}}$ put $x^n = t \rightarrow x = t^{1/n}$

$$dx = \frac{1}{n} t^{1/n-1} dt \rightarrow \begin{array}{c|c|c} x & 0 & 1 \\ \hline t & 0 & 1 \end{array}$$

$$I = \int_0^1 (1-t)^{-1/2} \frac{1}{n} t^{1/n-1} dt$$

$$\frac{1}{n} \int_0^1 t^{1/n-1} (1-t)^{1/2-1} dt$$

$$\frac{\pi}{2}$$

$$I = \frac{1}{n} B\left(\frac{1}{n}, \frac{1}{2}\right) \rightarrow \frac{\frac{1}{n} \Gamma\left(\frac{1}{2}\right) \Gamma\left(\frac{1}{2}\right)}{\Gamma\left(\frac{1}{n} + \frac{1}{2}\right)}$$

$$\rightarrow \frac{\sqrt{\pi}}{n} \frac{\Gamma\left(\frac{1}{n}\right)}{\Gamma\left(\frac{1}{n} + \frac{1}{2}\right)}$$

By DF, ~~in form~~ now multiply & divide by $\Gamma\left(\frac{1}{n}\right)$

$$\text{By DF, } \Gamma(n) \Gamma\left(\frac{n+1}{2}\right) = \frac{\sqrt{\pi}}{2^{2n-1}} \cdot (2n)! \quad \left(\text{here } \Gamma\left(\frac{1}{n}\right)\right)$$

$$\frac{\Gamma\left(\frac{1}{n}\right)}{\Gamma\left(\frac{1}{n} + \frac{1}{2}\right)} = \frac{\sqrt{\pi}}{2^{2n-1}} \cdot \frac{1}{\Gamma\left(\frac{1}{n}\right)}$$

$$I = \frac{\sqrt{\pi}}{n} \left(\frac{1}{n}\right)^2 \cdot 2^{2n-1} \rightarrow \frac{2^{2n-1} \left(\frac{1}{n}\right)^2}{n \sqrt{\frac{2}{n}}}$$

$$2. \int_0^1 \frac{x^{2n}}{\sqrt{1-x^2}} dx = \frac{(2n)! \pi}{2^{2n+1} (n!)^2}$$

$$I = \int_0^1 \frac{(x^2)^n}{\sqrt{1-x^2}} dx \rightarrow \text{put } x^2 = t, \quad x = \sqrt{t}$$

$$\rightarrow dx = \frac{1}{2\sqrt{t}} dt \rightarrow \begin{array}{c|c|c} x & 0 & 1 \\ \hline t & 0 & 1 \end{array}$$

$$I = \int_0^1 \frac{t^n}{\sqrt{1-t}} \cdot \frac{1}{2\sqrt{t}} dt \rightarrow \frac{1}{2} \int_0^1 t^{n-1/2} (1-t)^{1/2} dt$$

$$= \frac{1}{2} \int_0^1 x^{(n+1/2)-1} (1-x)^{1/2-1} dx$$

$$= \frac{1}{2} B(n+1/2, 1/2)$$

$$\rightarrow \frac{1}{2} \frac{\Gamma(n+1/2) \Gamma(1/2)}{\Gamma(n+1)} \rightarrow \frac{\sqrt{\pi}}{2} \frac{\Gamma(n+1/2)}{(n!)} \times \frac{\Gamma(n)}{\Gamma(n)}$$

where $\Gamma(n) \Gamma(n+1/2) = \frac{\sqrt{\pi}}{2^{2n-1}} \Gamma(2n)$

$$I = \frac{\sqrt{\pi} \sqrt{\pi} \sqrt{2n}}{2(n!) (n-1)! 2^{2n-1}}$$

$$\sqrt{2n} = (2n-1)!$$

$$\sqrt{n} = (n-1)!$$

$$\frac{\pi}{2^n (n!)} \frac{\pi (2n-1)!}{2^n (n!) (n-1)!} \times \frac{2n}{2n}$$

$$\frac{\pi (2n)!}{2^{2n+1} (n!)^2}$$

$$B(m, m) B(m+1/2, m+1/2) = \frac{\pi}{2} 2^{1-4m}$$

$$\frac{\Gamma(m) \Gamma(m)}{\Gamma(2m)} \times \frac{\Gamma(m+1/2) \Gamma(m+1/2)}{\Gamma(2m+1)}$$

Exercises 1-10
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$$\left[\frac{1}{2m} \frac{1}{(2m)} \frac{1}{2m} \right]^2 \rightarrow \frac{1}{2m} \left[\frac{1}{2m} \frac{1}{2m} \right]^2$$

Differentiation Under Integral Sign

DUIS

$$\frac{d}{dt} \int_a^b f(x, t) dx = \int_a^b \frac{\partial}{\partial t} f(x, t) dx$$

Rule of DUIS

$$I(t) = \int_0^{\pi} \frac{\sin(tx)}{x} dx \rightarrow I'(t) = \int_0^{\pi} \frac{\partial}{\partial t} \left[\frac{\sin(tx)}{x} \right] dx$$

$$= \int_0^{\pi} \cos(tx) \cdot x dx$$

Show that

① $\int_0^1 \frac{x^a - 1}{\log x} dx = \log(a+1) \quad (a > -1)$

② $\int_0^{\infty} \left[\frac{1 - \cos mx}{x} \right] e^{-x} dx = \frac{1}{2} \log(m^2 + 1)$

Differentiate
 it
 w.r.t. parameter
 & evaluate
 at
 the value of
 parameter
 & integrate

$(1+t) \log = (0) 1$

→ answer at the end of book.

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$$\textcircled{3} \int_0^{\infty} \left[\frac{1 - e^{-ax}}{x} \right] e^{-x} dx = \log(1+a)$$

Ans $\textcircled{1} \int \frac{x^a - 1}{\log x} dx = \log(a+1) \quad (a > -1)$

let $I(a) = \int \frac{x^a - 1}{\log x} dx$ — ① where a is the parameter.

Diff ① w.r.t a

$$I'(a) = \int \frac{d}{da} \left(\frac{x^a - 1}{\log x} \right) dx = \int \frac{d}{da} (x^a \cdot \log x) dx$$

$$= \int x^a \log x dx$$

$$\left[\frac{x^{a+1}}{a+1} \right]_0^1$$

$$I'(a) = \frac{1}{a+1} \rightarrow \text{let } a = t \quad I(a) = \int \frac{1}{a+1} da + c$$

$$\rightarrow I(a) = \log(a+1) + c \quad \text{--- ②}$$

put $a=0$ in ① & ②

$$\left. \begin{array}{l} \textcircled{1} \rightarrow I(0) = 0 \\ \textcircled{2} \rightarrow I(0) = 0 + c \end{array} \right\} \underline{\underline{c=0}}$$

$$\underline{\underline{I(a) = \log(a+1)}}$$

$$\textcircled{1} \int_0^{\infty} \left[\frac{1 - \cos mx}{x} \right] e^{-x} dx = \textcircled{1} \text{ integral}$$

where m is the parameter.

Diff w.r.t m , using DUIS

$$I'(m) = \int_0^{\infty} \frac{d}{dm} \left(\frac{1 - \cos mx}{x} \right) e^{-x} dx$$

$$= \int_0^{\infty} \frac{e^{-x}}{x} (x \sin mx) dx$$

imp formulae

$$\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$= \left[\frac{e^{-x}}{1+m^2} (-\sin mx - m \cos mx) \right]_0^{\infty}$$

$$0 - \frac{-1}{1+m^2} (0-m)$$

$$I'(m) = \frac{m}{1+m^2}$$

Integrate w.r.t. m

$$I(m) = \int \frac{m}{1+m^2} dm + c$$

$$\frac{d}{dx} \log(x) = \frac{1}{x}$$

Now we will multiply & divide by 2 to make $\frac{d}{dx} \log(x) = \frac{1}{x}$ form.

$$I(m) = \frac{1}{2} \int \frac{2m}{1+m^2} dm + c = \frac{1}{2} \log(1+m^2) + c$$

Parameter

now we will have to insert the value of m in main equation to see whether it comes zero or not. If it comes zero, then value of m is to be substituted in derived equations.

Let $x=0$ then $m=0$ = minimum

put $m=0$ in ① & ②

$$\left. \begin{array}{l} \text{①} \rightarrow I(0) = 0 \\ \text{②} \rightarrow I(0) = 0 + c \end{array} \right\} \rightarrow \underline{c=0}$$

Show that

$$\text{①} \int_0^{\infty} \frac{\sin(ax) e^{-bx}}{x} dx = \tan^{-1} \left(\frac{a}{b} \right)$$

$$\text{②} \int_0^1 \frac{x^a - x^b}{\log x} dx = \log \left(\frac{a+1}{b+1} \right)$$

③ $\int_0^{\infty} \frac{1 - \cos ax}{x^2} dx = \frac{\pi a}{2}$ given $\int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$

Ans ① $\int_0^{\infty} \frac{\sin(ax) e^{-bx}}{x} dx = \tan^{-1}\left(\frac{a}{b}\right)$

let b be the parameter:

$$I(b) = \int_0^{\infty} \frac{\sin(ax) e^{-bx}}{x} dx \quad \text{--- (1)}$$

$$= \int_0^{\infty} \frac{\sin(ax)}{x} \cdot e^{-bx} (-x) dx$$

$$= - \int_0^{\infty} e^{-bx} \sin(ax) dx$$

$$= - \int_0^{\infty} \frac{e^{-bx}}{b^2 + a^2} (-b \sin ax - a \cos ax) dx$$

$$\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} (a \sin bx - b \cos bx)$$

$$I'(b) = - \left\{ 0 - \frac{1}{b^2 + a^2} (0 - a) \right\}$$

$$I'(b) = \frac{-a}{a^2 + b^2} \quad \left| \int_0^{\infty} \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) \right|$$

$$I(b) = -a \int \frac{1}{a^2 + b^2} db + c$$

$$= -a \cdot \frac{1}{a} \tan^{-1}\left(\frac{b}{a}\right) + c \quad \text{--- (2)}$$

determinants more x² sign in integrals (1) & (2)
 because of larger of

put $b = \infty$ in (1) & (2)

$$\begin{aligned} (1) &\rightarrow I(\infty) = 0 \\ (2) &\rightarrow I(\infty) = \frac{\pi}{2} + C \end{aligned}$$

$$C = \frac{\pi}{2}$$

$$I(b) = \frac{\pi}{2} - \tan^{-1}\left(\frac{b}{a}\right) \rightarrow \tan^{-1}\left(\frac{a}{b}\right)$$

Show that

$$1. \int_0^{\infty} \frac{1 - \cos ax}{x^2} dx = \frac{\pi a}{2} \quad \text{given } \int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$$

$$2. \int_0^{\infty} \frac{\log(1 + ax^2)}{x^2} dx = \frac{\pi}{2} a$$

ordinary method se
 nhi ho yega jab
 denom mein x^2 ,
 log x aise aaye
 to

$$3. \int_0^1 \frac{x^a - x^b}{\log x} dx = \log \left(\frac{a+1}{b+1} \right)$$

1. Let $I(a) = \int_0^{\infty} \frac{1 - \cos ax}{x^2} dx$ (1)

Differentiate

$$I'(a) = \int_0^{\infty} \frac{\partial}{\partial a} \left(\frac{1 - \cos ax}{x^2} \right) dx$$

$$\int_0^{\infty} \frac{1}{x^2} [0 + (\sin ax) \cdot x] dx$$

$$I'(a) = \int_0^{\infty} \frac{\sin ax}{x} dx$$

$$\frac{x}{x} \left| \begin{matrix} 0 & \infty \\ 0 & \infty \end{matrix} \right|$$

put $ax = t \rightarrow dx = dt/a$

$$I'(a) = \int_0^{\infty} \frac{\sin t}{\left(\frac{t}{a}\right)} \cdot \frac{dt}{a} \rightarrow \frac{\pi}{2}$$

Integrate w.r.t a .

$$I(a) = \frac{\pi}{2} \int da + c \rightarrow \frac{\pi a}{2} + c$$

put $a=0$ in eqn ① & ②

$$\textcircled{1} \Rightarrow I(0) = 0 \quad \textcircled{2} \Rightarrow I(0) = c$$

$$I(a) = \frac{\pi a}{2} \quad \checkmark$$

2. $\int_0^{\infty} \frac{\log(1+x^2)}{(1+x^2)^2} dx = \pi \sqrt{a}$

$$\int \frac{1}{1+x^2} dx \rightarrow \int \frac{1}{x^2+a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$$

$$\frac{1}{a} \int \frac{1}{x^2 + \left(\frac{1}{a}\right)^2} dx$$

$$\frac{1}{a} (\frac{1}{\sqrt{a}}) \tan^{-1} (\frac{x}{\sqrt{a}})$$

$$\rightarrow \frac{1}{\sqrt{a}} \tan^{-1} (\frac{x}{\sqrt{a}})$$

3. $I(b) = \int_0^a \frac{x^a - x^b}{\log x} dx$ where a is parameter & b is constant.

$$I'(a) = \int_0^a \frac{\partial}{\partial a} \left(\frac{x^a - x^b}{\log x} \right) dx$$

$$\frac{d}{dx} (a^x) = a^x \log a$$

$$\int_0^a \frac{1}{\log x} [x^a \cdot \log x - 0] dx$$

$$\int_0^a x^a dx = \left[\frac{x^{a+1}}{a+1} \right]_0^a \quad I'(a) = \frac{1}{a+1}$$

Integrate w.r.t a :

$$I(a) = \int \frac{1}{a+1} da + c \rightarrow \log(a+1) + c$$

Put $a=b$ in ① & ②

$$\textcircled{1} \quad I(b) = 0$$

$$c = -\log(b+1)$$

$$\textcircled{2} \quad I(b) = \log(b+1) + c$$